Neel Shah

Computer Vision · Multi-Object Tracking · ML Engineering

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github.com/neeljshah/court-vision | linkedin.com/in/neeljshah22

SUMMARY

I build computer vision pipelines from the math up. My project, CourtVision, turns NBA broadcast video into player court coordinates on one consumer GPU: **YOLOv8** detection, **SIFT homography** hardened for broadcast footage, a **Kalman** and **Hungarian** tracker built from primitives, and **OSNet** re-ID reimplemented in **PyTorch**. I measured the output: about 10.5 percent of same-ID steps on a test clip were physically impossible and calibration did not generalize across footage, so the work is paused and documented.

SELECTED PROJECT: COURTVISION

Calibrated multi-sport forecasting engine, with the validation harness and paper-only execution controls as first-class components.
Evidence: github.com/neeljshah/court-vision/blob/master/docs/QUANT_BRIEF.md · github.com/neeljshah/court-vision/blob/master/EVIDENCE.md

Vision pipeline

- **Tracker from primitives:** 6D constant-velocity **Kalman** filter plus **Hungarian** assignment over a blended IoU and appearance cost, with a greedy fallback.
- **Ball detector:** fine-tuned a single-class **YOLOv8n** detector and exported it to **ONNX** and **TensorRT**.
- **Homography:** court calibration from classical CV (HSV masking, Hough lines, line-intersection corners) and a SIFT strategy hardened for broadcast with **inlier gating**, **EMA smoothing**, drift re-anchoring and replay or scene-cut suspension.
- **Re-ID: OSNet**-x0.25 reimplemented in **PyTorch** with a layered inference fallback (**TensorRT**, torchreid, standalone, MobileNetV2, HSV histograms); it ships with ImageNet-pretrained weights and the production appearance model is the HSV histogram.
- **Measured status:** tracking-derived features carry zero importance in the production prop models and the tracker output is not yet reliable enough to feed a model, so tracking is paused with the measurements documented.

Modeling, validation and systems

- **Modeling and data:** per-stat **XGBoost** with isotonic calibration; four de-vig methods from scratch incl. Shin (1992); a 151-file intelligence layer incl. a 291,625-pair player-vs-player matchup matrix; AI agents propose signals and the gate rejects them.
- **Harness:** expanding-window walk-forward with a per-fold date assertion, truncation-invariance leak tests, acceptance only when a change beats baseline on two independent corpora, and a published minimum detectable effect for every market-relative row.
- **Private engine:** FastAPI with ~99 endpoints across 12 routers, 9 long-running daemons under a watchdog, a 430-module codebase, ~7,400-test pytest suite, CI/CD, 5 Docker images; an MCP server (interface documented publicly) answers queries from AI agents with cited artifacts.
- **Evidence site:** one unified, CI-gated site at https://neeljshah.github.io/court-vision/, built by one person directing AI coding agents; performance-budget and accessibility (axe) gates are hard CI steps with zero serious or critical violations; a PAPER ONLY desk record with receipts.

PROFESSIONAL EXPERIENCE

Data Analyst Intern, Sunsolor · *Tempe, AZ · 2025*
Built live Tableau / Power BI suites; cut manual reporting latency ~30%; automated daily market-trend pipelines in Python; data-quality audits across 5+ datasets; optimized SQL on 1M+-row datasets.

Data Annotation Specialist, DataAnnotation.Tech · *Remote · 2024–2025*
Labeled 10,000+ structured samples for production ML training & LLM fine-tuning; authored corner-case documentation cutting pipeline error variance ~20%; refined labeling schemas with data scientists.

Business Intelligence Analyst, Fortrex Securities · *Madison, WI · 2023–2024*
Built secure executive reporting interfaces (99.9% uptime); refined backend payment APIs & reporting pipelines for high-throughput billing; engineered real-time anomaly-detection alerting.

EDUCATION & CERTIFICATIONS

University of Iowa: three years of data-science coursework. Coursework: math stats, probability theory, machine learning, financial analytics.
Certifications: Certified Data Scientist (USDSI) · ML Specialist & Data Viz (Kaggle) · CBIAP (IABAC).

SKILLS

ML/CV: PyTorch, OpenCV, YOLOv8/Ultralytics, ONNX, TensorRT, NumPy, Pandas, scikit-learn, XGBoost, LightGBM, statsmodels
Languages: Python (expert), SQL, JavaScript/TypeScript (working)
Backend/Infra: FastAPI, WebSocket/SSE, PostgreSQL, SQLite, Docker, GitHub Actions, GCP, Tableau, Power BI, dashboards
Methods: walk-forward / leak-free validation, calibration (isotonic), de-vigging (Shin), fractional-Kelly, Monte-Carlo simulation, shadow-logging, multi-agent orchestration